Optimization Frameworks for Hybrid Biomass Power-to-Methanol System: MILP and Reinforcement Learning Approaches

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Statement of Originality

I hereby certify that the work embodied in this thesis is the result of original research, is free of plagiarised materials, and has not been submitted for a higher degree to any other University or Institution.

Aug. 2018

Date

Signature

Muhammad Rafi Afif Indrajaya

Supervisor Declaration Statement

I have reviewed the content and presentation style of this thesis and declare it is free of plagiarism and of sufficient grammatical clarity to be examined. To the best of my knowledge, the research and writing are those of the candidate except as acknowledged in the Author Attribution Statement. I confirm that the investigations were conducted in accord with the ethics policies and integrity standards of Nanyang Technological University and that the research data are presented honestly and without prejudice.

Mar. 2019 — Signature

Date — Prof. Einstein XXX

Acknowledgements

I wish to express my greatest gratitude to my advisor.

“If I had one hour to save the world, I would spend 55 minutes defining the problem and only five minutes finding the solution.”

—Einstein, Albert

To my dear family

Abstract

My abstracts

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Symbols and Acronyms

Symbols

$\mathcal{R}^n$	the n-dimensional Euclidean space
$\mathcal{H}$	the Euclidean space
$\|\cdot\|$	the 2-norm of a vector or matrix in Euclidean space
$\|\cdot\|_G$	the induced norm of a vector in G-space
$\|\cdot\|_E$	the induced norm of a vector or matrix in probabilistic space
$\odot$	the Hadamard (component-wise) product
$\otimes$	the Kronecker product
$\langle \cdot, \cdot \rangle$	the inner product of two vectors
$\circ$	the composition of functions
∇f	the gradient vector
$\mathcal{C}^k$	the function with continuous partial derivatives up to k orders
$x_{i,k}$	the i-th component of a vector x at time k
$\bar{x}$	the vector with the average of all components of x as each element
$\mathbf{1}$	all-ones column vector with proper dimension
$\mathcal{C}$	the average space, i.e., $span\{\mathbf{1}\}$
$\mathcal{C}^\perp$	the disagreement space, i.e., $span^\perp\{\mathbf{1}\}$
$\Pi_\|$	the projection matrix to the average space $\mathcal{C}$
$\Pi_\perp$	the projection matrix to the disagreement space $\mathcal{C}^\perp$
$O(\cdot)$	order of magnitude or ergodic convergence rate (running average)
$o(\cdot)$	non-ergodic convergence rate
$\mathcal{N}_i$	the index set of the neighbors of agent i

Acronyms

DOP	Distributed Optimization Problem
EDOP	Equivalent Distributed Optimization Problem
SDOP	Stochastic Distributed Optimization Problem
OEP	Optimal Exchange Problem
OCP	Optimal Consensus Problem
DOCP	Dynamic Optimal Consensus Problem
AugDGM	Augmented Distributed Gradient Methods
AsynDGM	Asynchronous Distributed Gradient Methods
D-ESC	Distributed Extremum Seeking Control
D-SPA	Distributed Simultaneous Perturbation Approach
D-FBBS	Distributed Forward-Backward Bregman Splitting
ADMM	Alternating Direction Method of Multipliers
DSM	Distributed (Sub)gradient Method
GAS	Globally Asymptotically Stable
UGAS	Uniformly Globally Asymptotically Stable
SPAS	Semi-globally Practically Asymptotically Stable
USPAS	Uniformly Semi-globally Practically Asymptotically Stable
HoS	Heterogeneity of Stepsize
FPR	Fixed Point Residual
OBE	Objective Error
i.i.d.	independent and identically distributed
a.s.	almost sure convergence of a random sequence

Chapter 1

Introduction

1.1 Background

As hard-to-abate sectors advance their industrial-scale energy transition, Power-to-X solutions have become increasingly relevant. Yet these processes remain financially intensive because renewable energy availability is unpredictable and strongly influences both investment sizing and operational dispatch. In practice, MILP remains the robust industrial standard for optimizing such systems, while the broader rise of AI applications has motivated researchers to examine reinforcement learning as a decision-making agent for dispatch optimization. This development leads to the central methodological question of whether an RL-based approach can retain optimization quality while better representing operational complexity.

This question is especially relevant because MILP simplifies complex systems into linear approximations, which can reduce result accuracy when critical nonlinear behavior is present. A representative example is hydrogen tank operation, where the relationship between hydrogen level and efficiency is nonlinear because compression energy requirements increase at higher tank levels. Accordingly, this thesis investigates the effect of integrating this nonlinear tank behavior into the proposed system and evaluates its impact on the techno-economic results produced by the RL agent. Capturing this level of detail is not readily integrated into a MILP formulation and can become a significant limitation in complex polygeneration systems with nonlinear interactions across components and reactions.

Against this background, this thesis compares the techno-economic results produced by an RL agent with those from a conventional MILP framework for a hybrid biomass power-to-methanol plant in South Africa across different scenarios and case studies. The goal is to clarify the role of RL in complex Power-to-X industrial processes and benchmark its current performance against an industrial-standard framework.

1.2 Introduction to Company

This master's thesis research is conducted at FEV Europe GmbH in the energy+resources department. FEV Europe GmbH is a leading expert in the automotive industry, widely known for benchmarking and consulting services, and is now supporting innovation in the energy sector. Within this context, the thesis contributes to identifying an efficient optimization framework for innovative energy solutions in complex polygeneration plants for future FEV Europe GmbH projects.

1.3 Objective

This thesis systematically runs and analyzes two optimization frameworks under different scenarios to evaluate whether a nonlinear physical model with an RL agent yields significantly different component sizing and layout than MILP. It further assesses whether such differences translate into changes in techno-economic performance metrics, including CAPEX, OPEX, and LCOM. Based on the simulation outcomes, the study then identifies the reasons behind observed discrepancies, or the lack thereof, and determines whether one framework appears consistently better overall or whether framework suitability is scenario-dependent.

This comparative study is expected to provide a clearer understanding of performance differences between the industrial-standard MILP framework and the novel AI-based RL alternative. Building on this understanding, the thesis seeks to establish a more grounded perspective on the robustness and overall performance of each framework when applied to a complex polygeneration system.

1.4 Thesis Configuration

Chapter 2 reviews the state of the art in three areas: hybrid biomass power-to-methanol systems, MILP-based optimization strategies for complex energy systems, and recent RL applications for optimization in industrial plants and energy systems. Building on this foundation, Chapter 3 presents the methodology, including the selected hybrid biomass power-to-methanol system model, the Level 1 full-MILP reference approach, the Level 2 decoupled sizing and dispatch strategy using Bayesian optimization, and the Level 3 RL-based optimization strategy, together with the techno-economic parameters, scenarios, and KPIs considered. Chapter 4 then reports the benchmarking results for Levels 1, 2, and 3 across the defined scenarios and evaluates framework performance using the selected KPIs. Finally, Chapter 5 summarizes the main findings and discussion in line with the established thesis structure.

Chapter 2

Literature Review

2.1 Power-to-X Integration for Methanol Synthesis System

Different definitions of linearity and non-linearity for methanol synthesis systems are discussed in the literature.

2.2 MILP for Complex Polygeneration System

This section is kept as a placeholder and will be expanded with detailed MILP-specific discussion.

2.3 State-of-the-art RL in Optimizing Energy System and Industrial Processes

The adoption of RL to optimize energy systems and industrial processes has attracted significant attention due to flexibility and adaptability [1]. By directly interacting with the system, an RL agent derives strategies through a Markov Decision Process (MDP). This process involves dynamic evolution of states, actions, transition probabilities, and rewards at each time step.

- Diagram of MDP process.

The main objective of an agent is to maximize the discounted sum of reward from time t to infinity.

$$R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \tag{2.1}$$

The agent navigates actions using a policy π, which determines what to do at each state. Each policy has an associated state-value function $V^\pi(s)$.

$$V^\pi(s) = \mathbb{E}_\pi[R_t \mid s_t = s] \tag{2.2}$$

The optimal value function is defined over all policies.

$$V^*(s) = \max_\pi V^\pi(s) \tag{2.3}$$

Another related variable is the action-value function $Q^\pi(s, a)$.

$$Q^\pi(s, a) = \mathbb{E}_\pi[R_t \mid s_t = s, a_t = a] \tag{2.4}$$

Its optimal form is defined as follows.

$$Q^*(s, a) = \max_\pi Q^\pi(s, a) \tag{2.5}$$

In short, an optimal policy π^* is one whose state-value function equals the optimal state value.

$$\pi^* = \arg\max_\pi V^\pi(s) \tag{2.6}$$

- Diagram of these variables as an MDP process (to be recreated from literature).

Accurate environment dynamics modeling through MDPs is emphasized by many researchers to obtain robust frameworks and good results [2]. This supports RL use for complex and dynamic environments such as polygeneration plants.

- Examples of RL agents used in energy-system optimization.

Different RL algorithm families are commonly used for complex energy-system optimization: value-based, policy-based, and actor-critic. They differ mainly by learning mechanism. Value-based RL learns a value estimate, commonly $Q(s, a)$, then chooses actions with highest estimated value. Typical examples include Q-learning and DQN [3]. A related study argues value-function methods can face convergence issues in continuous multivariate settings, where they are less suitable than in discrete cases [4].

Policy-based RL learns the policy directly, parameterizing the action rule and searching for an optimal policy, including methods such as policy gradients and PPO [5]. Because many energy problems are continuous, high-dimensional, and constrained, policy-based approaches are often preferred [6].

Actor-critic methods combine both views: the actor learns policy and the critic learns value estimates to stabilize training. Earlier literature adopted Q-learning more frequently, while recent work increasingly applies actor-critic and other deep RL variants to complex control tasks [7]. Common algorithms include PPO, SAC, A3C, TD3, and DDPG.

2.4 RL vs MILP for Different Applications

Recent literature provides several RL-vs-MILP benchmarks for energy systems. Vetter et al. [8] compared DQN-based RL against MILP for dispatch optimization in a Grid-PV-TES energy community system with fixed sizing. RL controlled TES SOC, while MILP optimized detailed power flows. Both reduced annual cost versus baseline, with MILP showing larger cost reduction. RL, however, performed better on some additional metrics such as CO_2 reduction, self-consumption ratio, and self-sufficiency ratio because it tended to consume more PV generation.

Beyond dispatch-only benchmarking, Cauz et al. [9] compared RL and MILP under both design and control decisions in Battery-PV systems. The RL algorithm used was Direct Environment and Policy Search (DEPS). Design decisions were battery and PV sizing; control decisions were battery and grid dispatch. In stage one (one-week case), both approaches converged under fixed sizing. Under joint design-control optimization, RL initially appeared cheaper, but this was traced to modeling limitations (missing end-SOC consistency). After correcting assumptions, MILP became cheaper, highlighting the importance of matched assumptions for fair comparison.

In stage two (one-year case), RL remained more expensive than MILP under both fixed-sizing and joint design-control scenarios, while producing substantially different design choices. The study attributes this to multiple causes, including hyperparameter sensitivity and algorithm selection.

Chapter 3

Methodology

3.1 Hybrid Biomass Power-to-Methanol System Model

- Energy system definition.
- Technical parameters definition.
- Economical parameters definition.

3.2 Two-steps Outerloop & Innerloop Strategy

The methodology adopts a two-step strategy in which layout/sizing and operational dispatch decisions are decomposed into outerloop and innerloop problems to improve tractability and interpretability.

3.3 Level 2 Optimization Architecture (MILP)

- Component-level non-linear modeling logic.
- Operation logic.
- Modeling tool: `oemof.solph`.

- Solver: CBC.
- Objective function definition.

3.4 Level 3 Optimization Architecture (RL)

- Component-level linear modeling logic.
- Chosen RL algorithm.
- Operation logic (MDP).
- Integration of Bayesian Optimization (BO) and RL.
- Objective function definition (reward function).

3.5 Data

This section is reserved for dataset sources, preprocessing, assumptions, and scenario-specific inputs.

3.6 Comparison Stages & Scenario Definition

This comparison is conducted in two steps:

- **Outerloop comparison**
  - Innerloop operational decision is fixed as an `oemof` problem.
  - For outerloop (layout selection fed into innerloop), different tools are compared: `oemof` investment block (IB), Bayesian Optimization (BO), and Design of Experiment (DOE).
  - Objective: identify which approach gives the best layout under both linear and non-linear economies of scale.
  - Components to optimize: electricity supply and Power-to-X blocks.

- **Innerloop comparison**
  - Compare RL dispatch in a physics-based plant model against `oemof` dispatch in a linear model.
  - Stage 1 (fixed outerloop, linear model with RL): evaluate how close RL can optimize a linear model compared to Level 2 MILP.
  - Stage 2 (fixed outerloop, non-linear model with RL): evaluate behavior when optimized linear layouts are injected into a physics-based model, including feasibility and performance changes.
  - Stage 3 (joint outerloop + innerloop): compare BO+RL non-linear pathway vs BO+`oemof` linear pathway.

Case studies for each step (especially innerloop stage comparisons):

- **Weather variability**
  - Stable solar and wind input.
  - Extreme weather volatility.
- **Market conditions**
  - Cheap batteries.
  - Cheap hydrogen system (e.g., -50%, -20%, +20%, +50%).
- **System design and flexibility**
  - Heavy fixed methanol reactor load.
  - Heavy fixed electrolyzer load.

For each step, stage, and case study, predetermined KPIs are examined:

- **Techno-economic level**
  - LCOM.
  - CAPEX and OPEX.
  - Carbon intensity.

  - Grid dependency ratio.
  - Self-consumption ratio.
  - Autonomy days.

- **Framework level**
  - Simulation time.
  - BO distance.

- **Scenario descriptors**
  - Year.
  - CAPEX and OPEX per component.
  - Conversion factor per component.

- **Fixed components**
  - MeOH reactor.
  - Distillery.
  - MeOH storage.
  - Anaerobic digester.
  - ATR.
  - O_2 storage (full ATR-load sizing for 3 days).
  - Steam generator.
  - Auxiliary units: RWGS, fuel boiler, AD gas separator, compressor, and three heat exchangers.

- **Variable components (IB vs BO vs DOE)**
  - CH_4 generator or gas turbine.
  - Wind turbine.
  - PV.
  - BESS.
  - Electrolyzer.
  - Hydrogen storage.
  - Biogas storage.

Chapter 4

Results and Discussions

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Chapter 5

Conclusions

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5.1 Section 1

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5.2 Section 2

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Appendix A

Proofs for Part I or Chapter 3

A.1 Proof of Lemma

$$\psi^{av}(\theta) = \frac{1}{T}\int_0^T [\psi(\theta + \mu(\tau)) + C] \otimes \frac{\mu(\tau)}{a} d\tau$$

A.2 Proof of another Lemma

$$\begin{aligned} \gamma_1(\|x\|) \leq W(t,x) \leq \gamma_2(\|x\|) \\ \frac{\partial W}{\partial t} + \frac{\partial W}{\partial x}\phi(t,x,0) \leq -\gamma_3(\|x\|) \end{aligned} \tag{A.1}$$

List of Author's Awards, Patents, and Publications[1]

Awards

- **Best Paper Awards**, "A Great System," *Nature*.

Patents

- **A Great System**, "A Great System," *Nature*.

Journal Articles

- **My name** and My colleague, "A Great System," *Nature*.

Conference Proceedings

- **My name**, My colleague 1, My colleague 3 and My colleague 3, "Greater System," in *Conference of Vision, 2018*.

[1] The superscript * indicates joint first authors

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